Fiscal Year:	FY 2016	**Task Last Updated:**	FY 08/13/2015
PI Name:	Mao, Xiao Wen M.D.		
Project Title:	Space Flight Environment Induces Remodeling of Vascular Network and Glia-Vascular Communication in Mouse Retina		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	10/30/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gridley, Daila Ph.D. (Loma Linda University) Pecaut, Michael Ph.D. (Loma Linda University) Sun, Shu-Wei Ph.D. (Loma Linda University)		
Grant/Contract No.:	NNX15AB41G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The eye is a unique organ because it is relatively unprotected and constantly exposed to atmospheric oxygen, background radiation, environmental chemicals, and physical abrasion. A recent report shows that more than 30% of the astronauts returning from space were diagnosed with eye problems that can cause headaches and blurry vision. Our Preliminary studies from mice that had been subjected to space flights showed that environmental conditions during space travel lead to oxidative stress and induce adverse microvessel remodeling in the retina. However, our knowledge about the pathophysiological process from adaptive response to irreversible oxidative damage in the retina during and after space flight exposure and the underlying cellular mechanism(s) of these disturbances is very limited and inconsistent. The objectives of this application are to characterize the impact of space flight on female mouse retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and cell-cell interactions in the function of blood-retina-barrier (BRB). Our study will also determine the extent to which an antioxidant metalloporphyrin protects against space flight environment-induced changes in retinal neurovascular coupling and retinal function. The hypothesis will be tested in the specific aims with the spaceflight experiments on the International Space Station (ISS) and compared to that of ground-based groups using matured female mice. For flight studies, there will be two experiments over two flights, one for in-flight adaptation study on the ISS and one for studying re-adaptation after live animal return to Earth following long-term ISS mission. Selected groups of mice will be treated intraperitoneally with an antioxidant a week before launch and weekly during space flight. Animals will be sacrificed in orbit at 14 and 30 days after launch or 30 days and up to 6 months after return alive on Earth. Together, our unique, integrative, and quantitative activities with advanced MRI (magnetic resonance imaging) imaging techniques and retinal functional testing using visual electroretinography will provide insight into the molecular mechanisms and pathways of space flight condition-induced mitochondrial oxidative damage on retinal BRB integrity, vascular remodeling, and retinal function. The data will provide unique criteria for risks of functional detriments associated with space flight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Task Progress:	The objectives of this application are to characterize the impact of spaceflight on female mouse retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and cell-cell interactions in the function of blood-retina-barrier (BRB). Our study will also determine the extent to which an antioxidant metalloporphyrin protects against spaceflight environment-induced changes in retinal neurovascular coupling and retinal function. Because our flight experiment has not been scheduled and major funding has not been initiated, we currently have no data to present. In the meantime, we have worked with Paula Dumars at Ames Research Center to update the Flight Experiment Requirements Document (ERD). I also attended a rodent user group meeting at NASA Ames Research Center to present a summary of my experiment. NASA presented the current flight status and plans for live animal return. We met with project leaders and investigators to discuss the flight research projects. An antioxidant metalloporphyrin will be used in this study to determine if it can protect against spaceflight environment-induced changes in retinal neurovascular coupling and retinal function. To gain insight into the tissue distribution as well as to provide data that would ensure proper dosing regimens, we will conduct ground-based comprehensive pharmacokinetic study. Collaborators have been contacted to discuss the experiment design for the ground-based study.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Abstracts for Journals and Proceedings	Mao XW, Crapo J, Gridley D. "Radioprotection of Rat Eyes by a Superoxide Dismutase Mimic: a Metalloporphyrin Compound." Presented at 2015 Conference on Normal Tissue Radiation Effects and Countermeasures (CONTREC), Little Rock, AK, May 6-9, 2015. 2015 Conference on Normal Tissue Radiation Effects and Countermeasures (CONTREC), Little Rock, AK, May 6-9, 2015. , May-2015